Day 70 - Final Demo & Handover Report

Final system validation and project handover | 9 November 2026

Executive summary

The final local validation run exercised the Zecpath AI hiring pipeline from structured resume intake through ATS scoring, screening, interviews, machine-test scoring, unified scoring, and final recommendation. Four fictional profiles were processed and recorded in a JSON report. The repository's automated test suite also passed. This demonstrates an integrated local simulation; it does not represent a production deployment or a live company presentation.

Demo outcomes

- Alex Morgan: ATS 95.59; unified score 90.45; Selected.
- Riley Chen: ATS 94.71; unified score 81.44; Selected.
- Taylor Brooks: ATS 67.91; unified score 77.73; Selected.
- Jamie Rivera: ATS 70.60; unified score 61.71; Hold / Review.

All profiles and assessment values are synthetic fixtures. The recorded pipeline time was 0.7726 seconds. Model initialization and any first-use download are excluded, so this figure is not a production benchmark.

Finding for review

HR scores ranged from 87.93 to 90.12 across profiles with different experience and technical results. The narrow 2.19-point spread indicates that the current deterministic HR heuristics do not distinguish these examples strongly. HR output should be treated as a conversation aid rather than a reliable ranking signal.

Completion status

The local demo, written handover materials, and final submission files are prepared. The live demonstration to the company, a team knowledge-transfer session, and the internship evaluation remain pending. No external evaluator feedback or score is included.

System architecture and scoring

The orchestration entry point is `main_day63_end_to_end_simulation.py`. It reads a synthetic-only manifest from `data/day63_demo/`, processes each profile, and writes stage results, recommendations, and known limitations to a JSON report.

Pipeline stages

- Resume intake uses pre-structured sample JSON; the run does not exercise file upload or resume-parser integration.
- The ATS service normalizes the resume, masks protected fields on a scoring copy, extracts skills, and calculates experience, education, and semantic relevance.
- Screening uses the conversation state machine and a configured rubric to score supplied sample answers.
- The HR interview runs a scripted flow with fictional responses and deterministic demonstration heuristics.
- The technical interview stores sample responses, while its numeric score is supplied by a rubric fixture.
- The machine-test record contains sample code and a simulated execution summary; the code is not executed or sandboxed.
- Unified scoring and eligibility rules feed the final recommendation engine and report.

Semantic model

The matching engine loads the Sentence Transformers model `all-MiniLM-L6-v2`. It converts text into embeddings and compares them with cosine similarity; recent embeddings are cached. This supports semantic comparison between resume language, listed skills, and job requirements. It is an embedding model, not a generative chat model, and the Day 70 harness does not call a generative LLM.

Scoring logic

For the Python Developer demo role, the default ATS weights are skills 35%, experience 30%, education 15%, and semantic similarity 20%. Component values are normalized from 0 to 1 before conversion to a percentage.

The unified score combines five scores from 0 to 100: ATS 25%, screening 15%, HR 20%, technical interview 25%, and machine test 15%. Default interview rubrics are HR relevance 30%, communication 25%, confidence 20%, consistency 25%; technical accuracy 40%, depth 25%, reasoning 20%, applicability 15%; and machine-test correctness 40%, efficiency 20%, code quality 20%, problem solving 20%.

Recommendation defaults are Selected at 75 or above, Hold / Review from 50 to below 75, and Rejected below 50. Eligibility failures or review risks can override score thresholds. Human review remains necessary for any real hiring decision.

Validation, handover, and next steps

Verification

The full automated test suite completed with 679 passed and 8 subtests passed. The Day 70 run is saved at `reports/day70_final_demo.json`. Its reported pipeline duration covers the measured stage execution only and should not be used as a production service-level claim.

Reproducing the local demo

From PowerShell at the repository root, activate the virtual environment and run the command below. It saves a separate output file and preserves the recorded Day 70 run:

```
.\venv\Scripts\Activate.ps1
python -c "from pathlib import Path; from main_day63_end_to_end_simulation import
run_simulation; result =
run_simulation(output_path=Path('reports/day70_demo_rerun.json'));
print('Profiles:', result['candidate_count'])"
```

Current boundaries

This is a local prototype using synthetic candidate records, prewritten interview answers, and rubric fixtures. It does not connect to the company website, provide a production UI or API, verify real candidate consent, automatically evaluate the technical answers, or execute submitted code. Real hiring use would require additional validation, suitable controls, and human review.

Handover and evaluation status

Prepared materials include the local demo report, architecture and scoring notes, rerun instructions, and the final handover PDF. The live company demonstration, team knowledge-transfer session, and evaluator discussion still require external participation. The internship evaluation remains pending until genuine feedback is received.

Supporting files: `docs/day70/FINAL_HANDOVER.md` contains the extended code map and setup notes; `docs/day70/README.md` indexes the Day 70 materials.